INF10025 Data Management and Analytics

Tutorial 9 – Discussion Questions for Week 9

1. How many columns will be in the result set when the following statement is executed?

```
SELECT      M.title, M.mvyear, A.surname || ' ' || a.firstname As "Actor Name"
FROM        movie m
INNER JOIN  casting c
ON          c.movieno = m.movieno
INNER JOIN  actor a
ON          c.actorno = a.actorno
ORDER BY    1,3
```

2. What is a view?

3. What occurs when the following statement is executed?

```
CREATE VIEW mview AS
SELECT      M.title, M.mvyear, A.surname || ' ' || a.firstname As "Actor Name"
FROM        movie m
INNER JOIN  casting c
ON          c.movieno = m.movieno
INNER JOIN  actor a
ON          c.actorno = a.actorno
ORDER BY    1,3
```

4. How many columns will be in the result when the following statement is executed?

```
SELECT * FROM mview;
```

5. What will occur when this statement is executed?

```
SELECT * FROM mview
WHERE mvyear = 1998;
```

6. List 3 reasons why views are useful.

7. Does a view store any data?

8. What statement is used to delete the view named mview?

9. Can you insert rows by using a view name instead of a table name?

10. Can you delete rows by using a view name?

11. You have been asked to model housing information for the local government
Each house (which has a unique address) is owned by one or more owners. Each owner 'owns' a percentage of the house (e.g. Person A has 30% ownership while Person B has 70% ownership). Each house has a number of bedrooms. Each house has a number of bathrooms. Each owner may own more than one house. Each owner is identified by name and also has a phone number.

Each house may be rented by one or more renters. A renter is identified by name and also has a mobile phone no.

Some houses are not rented at all.

Draw an ERD diagram based on the above narrative.
Convert the ERD to a relation schema
Show examples of data stored in all the tables generated by your ERD

12. EatMoreFood is a company that makes food recipes available to customers via an on-line service.
Each recipe requires a number of ingredients. Each recipe is written by one or more authors.

While some recipes may have the same title as another recipe, each recipe submitted has a unique id. Authors may submit any number of recipes. Each ingredient used in a recipe will be stored within the database. Each ingredient will be available from one or more on-line suppliers. Details about each supplier will also be stored.

To use the system, a customer must first register their name and email address with EatMoreFood. A unique customer number will be assigned to each customer. Whenever a customer downloads a recipe, the system must store the date that the recipe was downloaded for that customer. If a recipe is downloaded more than once, then the system must record the date of each individual download.

Below are some examples of information that is currently stored by EastMoreFood in hand-written documents.

List of recipes:

Recipe No	*123456*	***Recipe Name***	*Baked Beans & Chili on Toast*
Recipe No	*666111*	***Recipe Name***	*Fried Tomato & Egg on Toast*
Recipe No	*342572*	***Recipe Name***	*Honey & Soy Chicken Breast*

Recipe Examples:

Recipe No	*123456*	
Recipe Name	*Baked Beans & Chili on Toast*	
Author(s) & contact phone numbers	*Fred Smith* *9123-4567*	*John Davies* *0419-999999*
Ingredient No & Name	***Qty***	
X100 White Bread	*2 slices*	
X201 Baked Beans	*1 750g Tin*	
X222 Karl's Hot Chilli Sauce	*1 teaspoon*	
X075 Butter	*1 tablespoon*	
Instructions:		
Pour beans onto plate and place in Microwave. Heat on High for 2 minutes. Toast bread then butter each slice. Pour beans over toast and sprinkle with Karl's Hot Chili sauce. Serve hot.		
Download Details		

Customer Name	***Email***	***Date***
Ted Brown	*tbrown@gmail.com*	*10 June*
Jenny Townsend	*JT01@hotmail.com*	*15 June*

Ingredient	*X100 White Bread*
Available from the following On-Line Suppliers:	
Supplier Code	***Supplier Name***
L001	*Langdon Supermarket*
S002	*Saveway Supermarket*
C006	*Coals Supermarket*

Recipe No	*666111*	
Recipe Name	*Fried Tomato & Egg on Toast*	
Author(s) & contact phone numbers	*Jenny Hill* *9123-4566*	
Ingredient No & Name	***Qty***	
X100 White Bread	*1 slices*	
X123 Egg	*1*	
X075 Butter	*1 teaspoon*	
X414 Tomato	*1*	
Instructions:		
Cut Tomato in half a fry in fry pan. Fry egg in fry pan at same time. Toast bread then butter it. Place fried egg and tomato on toast. Serve hot.		
Download Details		
Customer Name	***Email***	***Date***
Jason Black	*jassy@gmail.com*	*17 June*

Ingredient No & Name	*X222 Karl's Hot Chili Sauce*
Available from the following On-Line Suppliers:	
Supplier Code	***Supplier Name***
L001	*Langdon Supermarkets*
G002	*Godfrey's Delicatessen*

On-Line Supplier No	***On-Line Supplier Name***	***On-Line Supplier web address***
L001	*Langdon's Supermarket*	*www.langon_sm.com.au*
S002	*Saveway Supermarket*	*www.saveway_sm.com.au*
C006	*Coals Supermarket*	*www.coals_sm.com.au*
G002	*Godfrey's Delicatessen*	*www.godfreys_sm.com.au*

Requirements: Build an ERD to solve this problem.
If you don't know where to start, the next page has some tips / hints.

While the above problem seems lengthy, building the ERD is reasonably straight forward

This is one strategy for developing the ERD.

a. Determine the attributes of an Ingredient.

b. Determine the attributes of a Recipe.

c. Draw the Strong Entities for these entities

d. Decide if there is a relationship between these entities?
   If so then
   - Draw the relationship
   - **Read the relationship out aloud**. Does it make sense?
   - If the relationship is M:M then expand the relationship.
     Do any attributes belong to the intersection entity?
     - If so, add the attributes.

e. Determine the attributes of a Supplier.

f. Decide if there is a relationship between Supplier and Ingredient?
   If so then
   - Draw the relationship
   - **Read the relationship out aloud.** Does it make sense?
   - If the relationship is M:M then expand the relationship.
     Do any attributes belong to the intersection entity?
     - If so, add the attributes.

g. Determine the attributes of an Author.

h. Is there a relationship between Author and any of the other entities?
   If so then
   - Draw the relationship
   - **Read the relationship out aloud.** Does it make sense?
   - If the relationship is M:M then expand the relationship.
     Do any attributes belong to the intersection entity?
     - If so, add the attributes.

i. Determine the attributes of a Customer.

j. Is there a relationship between Customer and any of the other Entities?
   If so then
   - Draw the relationship
   - **Read the relationship out aloud.** Does it make sense?
   - If the relationship is M:M then expand the relationship.
     Do any attributes belong to the intersection entity?
     - If so, add the attributes.

Things to check:

- All strong entities must have their Identifiers underlined.
- All weak entities have their partial identifiers underlined
  PLUS underline the appropriate relationship names. **This step is very important.**

13. Burwood Rd College offers many subjects in its IT degree. Each subject is identified by a subject code and has a title and description. The code, title and description do not change over the life of a subject. A subject may be run over many semesters and years (e,g, Semester 1 2011, Semester 2 2011...).

Each semester, a subject assigned one Lecturer. That lecturer can change from semester to semester. Each Lecturer has a Lecturer ID and a Name. A lecturer may lecture in more than one subject in a single semester.
The College has many students. Each student has an Id and Name. Each student may enrol in one or more subject offerings. After completing a subject and its various assessment items, the students is awarded a Grade for that subject. (E.g. HD, P, N etc).
During a semester, a student may post comments to an online discussion board. Only students who are enrolled in a subject are eligible to post discussion board comments. The college is required to keep track of each comment and the date/time it was made. A student may make many comments in any subject he / she is enrolled in.

Requirements:

- Draw the ERD.
- Convert the ERD into a relation schema. Show all PKs and FKs.